Glivenko's theorem underneath structure

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Abstract. Glivenko's theorem has been studied in an abstract setting, where double negation is replaced by arbitrary nuclei, and classical and intuitionistic propositional logics are generalised to abstract entailment relations. This paper extends that framework to substructural logics, focusing on cases where structural rules such as weakening, contraction, and exchange are restricted. We apply our findings to substructural Full Lambek logic. All results are developed within an intuitionistic metatheory, allowing a computational interpretation and direct formalisation in Agda.

Keywords: Glivenko's theorem · Full Lambek calculus · Substructural logics · Nuclei · Structural proof theory · Agda

1 Introduction

Glivenko's theorem is one of the earliest and most influential results relating classical and intuitionistic logic. Established in the setting of propositional logic, it states that the classical provability of a formula implies the intuitionistic provability of its double negation [13,14]:

$$\mathbf{Cla} \vdash U \rhd a \iff \mathbf{Int} \vdash U \rhd \neg\neg a,$$

where **Cla** and **Int** denote classical and intuitionistic propositional logic, respectively.

Results of this kind are often referred to as *conservation* or *translation* results between logical systems. Their importance lies in the fact that they allow classical reasoning to be studied within a constructive framework. Rather than treating classical and intuitionistic mathematics as entirely separate enterprises, such results make it possible to identify which parts of classical reasoning can be faithfully interpreted in constructive terms. This is particularly significant because intuitionistic proofs often admit computational interpretations—for instance through the Curry–Howard correspondence—so that translating classical

arguments into intuitionistic ones may reveal their underlying computational content.

Over time, a number of generalisations of Glivenko's theorem have been investigated, often by replacing **Int** with weaker logics. In 1968, Segerberg [34] introduced the *Glivenko logic* **Gli**, defined as Johansson's minimal logic extended by the double negation of *ex falso quodlibet sequitur*, namely $\neg\neg(\bot \to a)$. In the same paper, he claimed—without proof or citation—a characterisation result that was later formally established by Odintsov [26,27]: the Glivenko logic is the smallest logic for which Glivenko's theorem holds. More precisely, for any logic $\mathbf{L}$ such that $\mathbf{Min} \subseteq \mathbf{L} \subseteq \mathbf{Cla}$, the following are equivalent:

1. $\mathbf{Cla} \vdash U \rhd a$ if and only if $\mathbf{L} \vdash U \rhd \neg\neg a$;
2. $\mathbf{L}$ is an extension of **Gli**.

Extensions of this result to substructural logics were developed by Cignoli and Torrens [6,7], and by Galatos and Ono [12,11]. Subsequently, Ono [29] established an analogue of Segerberg's characterisation for the Full Lambek logic with exchange, $\mathbf{FL_e}$—that is, intuitionistic linear logic without exponentials—and its *involutive* extension $\mathbf{InFL_e}$, obtained by adding the stability axiom $\neg\neg a \rhd a$ to $\mathbf{FL_e}$. Ono's theorem states that for any logic $\mathbf{L}$ such that $\mathbf{FL_e} \subseteq \mathbf{L} \subseteq \mathbf{InFL_e}$, the following conditions are equivalent:

1. $\mathbf{InFL_e} \vdash U \rhd a$ if and only if $\mathbf{L} \vdash U \rhd \neg\neg a$;
2. The following conditions are satisfied:
   - $\mathbf{L} \vdash \neg(a \wedge b) \rhd \neg(\neg\neg a \wedge \neg\neg b)$;
   - $\mathbf{L} \vdash \neg(a \to b) \rhd \neg(\neg\neg a \to \neg\neg b)$.

We refer to any logic satisfying these conditions as the *Glivenko extension* of $\mathbf{FL_e}$.

Double negation in intuitionistic logic is a specific instance of a *nucleus* [2,16,17,24,32,35,36,4]—also referred to as a strong monad [19] or lax modality [1]—defined as a map j on formulae satisfying:

$$(a \to j(b)) \lhd\!\rhd (j(a) \to j(b)).$$

Fellin and Schuster [9,10] generalised Glivenko's theorem by replacing double negation with an arbitrary nucleus, thereby moving from syntactic provability in a particular calculus to the more general setting of inductively generated abstract consequence relations, and extending the framework from propositional formulas to arbitrary objects. They provided precise criteria under which a classical-style generalisation yields a conservative extension of its intuitionistic counterpart via a nucleus.

The present work extends the framework of [9,10] to entailment relations that may lack structural rules such as weakening, contraction, and exchange [30,31]. In the traditional structural setting, the presence of exchange together with implication ensures that j remains a nucleus over all inductive extensions of $\mathbf{L}$, including the Glivenko extension. In the more general substructural context, however, this property no longer holds automatically. As a consequence, the

assumption that j is a nucleus over the initial relation cannot in general be propagated to its extensions, and the role of the nucleus must be analysed more carefully. This makes it possible, in particular, to dispense with the initial nucleus assumption that is typically required in the structural case.

Another important feature of the main theorem in the present paper is that it further investigates the inclusion relationships between the various extensions induced by nuclei, thereby generalising and expanding characterisations that have appeared in the literature.

In the final section, we apply our results to the substructural *Full Lambek logic* **FL** and its extensions [20,21,22]. For the commutative case, using the Glivenko nucleus, $\neg\neg$ [4,8], we recover the aforementioned results in [12,11,29]. We also extend our findings to substructural variants of the *open nucleus*, which remain meaningful even in the noncommutative setting.

All the results in this paper are developed within an intuitionistic metatheory. Working constructively at the meta-level ensures that the arguments themselves admit a computational interpretation and avoids relying on classical principles that would be incompatible with the constructive perspective underlying much of the work. An additional benefit of this approach is that the main constructions and proofs can be directly formalised in a proof assistant. In particular, we have implemented the results in Agda [25][4], providing a machine-checked formalisation of the main definitions and theorems. The source code is available at [5]. This formalisation is partially based on the Agda development by Altenkirch [3]. Throughout the paper, the symbol ↺ next to a definition or theorem header is a clickable link to the corresponding formalised statement in the HTML rendering of the Agda source.

2 Substructural entailment relations and conservation

Let S be a set and S^* be the set of finite lists of elements of S. Given $\mathbf{L} \subseteq S^* \times S$, we write $\mathbf{L} \vdash U \rhd b$ in place of $(U, b) \in \mathbf{L}$. The expression "$U \rhd b$" is called *sequent*.

A *(single-conclusion) substructural entailment relation* is a relation $\mathbf{L} \subseteq S^* \times S$ such that

$$\frac{}{a \rhd a}\,\text{Refl} \qquad\qquad \frac{U \rhd a \qquad V_1, a, V_2 \rhd b}{V_1, U, V_2 \rhd b}\,\text{Trans}$$

These two are the *structural rules* of *reflexivity* and *transitivity*, respectively.

Optional structural rules include *commutativity* (or *exchange*), *monotonicity* (or *weakening*) and *contraction*:

$$\frac{U_1, a_1, a_2, U_2 \rhd b}{U_1, a_2, a_1, U_2 \rhd b}\,\text{Comm} \qquad \frac{U_1, U_2 \rhd b}{U_1, a, U_2 \rhd b}\,\text{Mono} \qquad \frac{U_1, a, a, U_2 \rhd b}{U_1, a, U_2 \rhd b}\,\text{Contr}$$

Quite often an entailment relation $\mathbf{L}$ is inductively generated from a given set $\mathcal{R}$ of rules by closing up with respect to the structural rules above. These

[4] The formalisation is built on top of the `agda/cubical` library [37].

non-structural rules have the form

$$\frac{U_1 \rhd b_1 \quad \dots \quad U_n \rhd b_n}{U \rhd b}\, r \qquad \text{or, more compactly,} \qquad \frac{\{U_i \rhd b_i \colon 1 \leq i \leq n\}}{U \rhd b}\, r$$

We call *axioms* the rules that are premissless, i.e. have the form

$$\frac{}{U \rhd b}\, r$$

If $\mathbf{L}$ is inductively generated by a set $\mathcal{R}$ of rules, we write $\mathbf{L} = \langle \mathcal{R} \rangle$.

An inductively generated entailment relation $\rhd$ can have multiple generating rule sets $\mathcal{R}$. Despite this, it is common to refer to an inductively generated entailment relation $\rhd$ without specifying a set of generating rules $\mathcal{R}$, unless statements about $\rhd$ depend on $\mathcal{R}$.

By an *extension* $\mathbf{L}'$ of an entailment relation $\mathbf{L}$ we mean in general an entailment relation such that $\mathbf{L} \subseteq \mathbf{L}'$. Given an inductively generated entailment relation $\mathbf{L} = \langle \mathcal{R} \rangle$ and a set of rules $\mathcal{R}'$, then we call $\mathbf{L} + \mathcal{R}'$ the entailment relation inductively generated by $\mathcal{R} \cup \mathcal{R}'$. We also say that $\mathbf{L} + \mathcal{R}'$ is an *inductive extension* of $\mathbf{L}$, and we call $\mathcal{R}'$ the set of *additional rules*. Notice that this notation is somewhat informal since $\mathbf{L}$ does not determine $\mathcal{R}$.

Definition 1 ( ). *Consider the following rules:*

$$\frac{U, a, V \rhd j(b)}{U, j(a), V \rhd j(b)}\, \mathrm{L}j \qquad\qquad \frac{U \rhd b}{U \rhd j(b)}\, \mathrm{R}j$$

Given a set S endowed with an entailment relation $\mathbf{L}$, we say that a function $j \colon S \to S$ is

- expansive (over $\mathbf{L}$) *if* Rj *holds in* $\mathbf{L}$*;*
- *a* bi-nucleus (over $\mathbf{L}$) *if both* Lj *and* Rj *hold in* $\mathbf{L}$*.*

Definition 2 ( ). *Let $j \colon S \to S$. Given a rule r, its j-*version *r_j is obtained by putting j in front of every succedent:*

$$\frac{\{U_i \rhd b_i \colon i \in I\}}{U \rhd b}\, r \quad \rightsquigarrow \quad \frac{\{U_i \rhd j(b_i) \colon i \in I\}}{U \rhd j(b)}\, r_j$$

We write $\mathcal{R}_j$ for $\{r_j \colon r \in \mathcal{R}\}$. Given a rule r that is admissible in $\mathbf{L}$, we say that r survives after j (over $\mathbf{L}$) *if r_j is admissible in $\mathbf{L}$. We also define the* stability axiom *for j:*

$$\frac{}{j(a) \rhd a}\, j\text{-stab}$$

Remark 1 ( ). We observe the following (for more details, see Appendix A):

1. Rj and Refl_j are equivalent in $\mathbf{L}$.
2. Lj and Trans_j are equivalent in $\mathbf{L}$.

3. j-stab and the following are equivalent in $\mathbf{L}$

$$\frac{U, a, V \rhd b}{U, j(a), V \rhd b}\,\mathrm{L}j^{+}$$

Definition 3 ($\circlearrowleft$**).** *Given an inductively generated entailment relation* $\mathbf{L} = \langle \mathcal{R} \rangle$ *and an expansive function* j *on* $\mathbf{L}$*, we define the following extensions:*

1. *the* Kleisli j-extension $\mathrm{K}_j(\mathbf{L})$ *of* $\mathbf{L}$ *by*

$$\mathrm{K}_j(\mathbf{L}) := \{(U, a) \colon \mathbf{L} \vdash U \rhd j(a)\},$$

2. *the* Glivenko j-extension $\mathrm{G}_j(\mathbf{L})$ *of* $\mathbf{L}$ *as*

$$\mathrm{G}_j(\mathbf{L}) := \mathbf{L} + \{\mathrm{L}j\} + \mathcal{R}_j,$$

3. *the* Kleisli-stable j-extension $\mathrm{M}_j(\mathbf{L})$ *of* $\mathbf{L}$ *as*[5]

$$\mathrm{M}_j(\mathbf{L}) := \mathbf{L} + \{j\text{-stab}\}.$$

Proposition 1 ($\circlearrowleft$**).** *Let* S *be a set endowed with an inductively generated entailment relation* $\mathbf{L} = \langle \mathcal{R} \rangle$*, and let* j *be expansive over* $\mathbf{L}$*. Then the following inclusions hold:*

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[d, hook] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{M}_j(\mathbf{L})
\end{tikzcd}$$

Proof.

- $\mathbf{L} \subseteq \mathrm{K}_j(\mathbf{L})$ is nothing but $\mathrm{R}j$.
- $\mathbf{L} \subseteq \mathrm{G}_j(\mathbf{L})$ follows from the definition of $\mathrm{G}_j(\mathbf{L})$.
- $\mathrm{G}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L})$: It suffices to show that the latter admits all rules in the inductive definition of the former. The only non-trivial ones are the r_j where $r \in \mathcal{R}$. Let us consider

$$\frac{\{U_i \rhd b_i \colon i \in I\}}{U \rhd b}\,r \quad \rightsquigarrow \quad \frac{\{U_i \rhd j(b_i) \colon i \in I\}}{U \rhd j(b)}\,r_j$$

We have the following in $\mathrm{M}_j(\mathbf{L})$:

$$\dfrac{\dfrac{\dots \quad \dfrac{U_i \rhd j(b_i) \quad \dfrac{}{j(b_i) \rhd b_i}\,j\text{-stab}}{U_i \rhd b_i}\,\mathrm{Trans} \quad \dots}{U \rhd b}\,r}{U \rhd j(b)}\,\mathrm{R}_j$$

[5] The notation here is adapted from [11, §8.2]. The "M" stays for "maximum," as $\mathrm{M}_j(\mathbf{L})$ is the greatest extension $\mathbf{L}'$ of $\mathbf{L}$ such that $\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L})$. See Theorem 1.3, cf. [11, Theorem 8.7].

– $\mathrm{K}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L})$:

$$\dfrac{\dfrac{\dfrac{\mathrm{K}_j(\mathbf{L}) \vdash U \rhd b}{\mathbf{L} \vdash U \rhd j(b)}\,\text{defn}}{\mathrm{M}_j(\mathbf{L}) \vdash U \rhd j(b)}\,\text{defn} \qquad \dfrac{}{\mathrm{M}_j(\mathbf{L}) \vdash j(b) \rhd b}\,j\text{-stab}}{\mathrm{M}_j(\mathbf{L}) \vdash U \rhd b}\,\text{Trans}$$

□

Theorem 1 (Conservation). *Let* $\mathbf{L} = \langle \mathcal{R} \rangle$*, let* $\mathbf{L}' = \mathbf{L} + \mathcal{R}'$*, and let* j *be expansive over* $\mathbf{L}$*. Then:*

$$\mathrm{K}_j(\mathbf{L}') \supseteq \mathbf{L}' \tag{1}$$

$$\mathrm{K}_j(\mathbf{L}') \subseteq \mathbf{L}' \iff \mathbf{L}' \supseteq \mathrm{M}_j(\mathbf{L}) \tag{2}$$

$$\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L}) \iff \mathbf{L}' \subseteq \mathrm{M}_j(\mathbf{L}) \tag{3}$$

$$\mathrm{K}_j(\mathbf{L}') \supseteq \mathrm{M}_j(\mathbf{L}) \iff \mathbf{L}' \supseteq \mathrm{G}_j(\mathbf{L}) \tag{4}$$

Proof.

(1) A particular case of Proposition 1.

(2) If $\mathbf{L}' \supseteq \mathrm{M}_j(\mathbf{L})$, then in particular $\mathbf{L}' \vdash j(a) \rhd a$. Let us suppose $\mathrm{K}_j(\mathbf{L}') \vdash U \rhd a$, i.e. $\mathbf{L}' \vdash U \rhd j(a)$. Then by Trans we conclude $\mathbf{L}' \vdash U \rhd a$. We have shown $\mathrm{K}_j(\mathbf{L}') \subseteq \mathbf{L}'$.
Now suppose that $\mathrm{K}_j(\mathbf{L}') \subseteq \mathbf{L}'$. From $\mathbf{L}' \vdash j(a) \rhd j(a)$, i.e. $\mathrm{K}_j(\mathbf{L}') \vdash j(a) \rhd a$, we get $\mathbf{L}' \vdash j(a) \rhd a$ by assumption. This means that j-stab holds in $\mathbf{L}'$, which suffices for $\mathbf{L}' \supseteq \mathrm{M}_j(\mathbf{L})$.

(3) The direction "$\Rightarrow$" easily follows from $\mathbf{L}' \subseteq \mathrm{K}_j(\mathbf{L}')$ (Proposition 1).
Now suppose that $\mathbf{L}' \subseteq \mathrm{M}_j(\mathbf{L})$. Since $\mathbf{L} \subseteq \mathbf{L}'$, we have $\mathrm{M}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L}')$. On the other hand, as stability and all the rules of $\mathbf{L}'$ are admissible in $\mathrm{M}_j(\mathbf{L})$, we also have $\mathrm{M}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L})$. Therefore $\mathrm{M}_j(\mathbf{L}) = \mathrm{M}_j(\mathbf{L}')$. From this and $\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L}')$ (Proposition 1), we conclude $\mathrm{K}_j(\mathbf{L}') \subseteq \mathrm{M}_j(\mathbf{L})$.

(4) Assume $\mathbf{L}' \supseteq \mathrm{G}_j(\mathbf{L})$, and suppose $\mathrm{M}_j(\mathbf{L}) \vdash U \rhd a$. We prove that $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd ja$ by induction on the last rule r used in the derivation of $\mathrm{M}_j(\mathbf{L}) \vdash U \rhd a$. The case in which $r \in \mathcal{R} \cup \{\text{Trans}, \text{Refl}\}$ is converted in $\mathrm{G}_j(\mathbf{L})$ to an application of the rule r_j, while the case in which r is j-stab is converted to an instance of Refl. So we have proved $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd ja$, which by hypothesis implies $\mathbf{L}' \vdash U \rhd ja$, i.e. $\mathrm{K}_j(\mathbf{L}') \vdash U \rhd a$ as desired.
Assume $\mathrm{K}_j(\mathbf{L}') \supseteq \mathrm{M}_j(\mathbf{L})$. Suppose that $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd a$. We prove $\mathbf{L}' \vdash U \rhd a$ by induction on the last rule used in the derivation of $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd a$.
(a) Case $r \in \mathcal{R}$: We have

$$\dfrac{\mathrm{G}_j(\mathbf{L}) \vdash U_1 \rhd a_1 \quad \ldots \quad \mathrm{G}_j(\mathbf{L}) \vdash U_n \rhd a_n}{\mathrm{G}_j(\mathbf{L}) \vdash U \rhd a}\,r$$

which leads to

$$\dfrac{\dfrac{\mathrm{G}_j(\mathbf{L}) \vdash U_1 \rhd a_1}{\mathbf{L}' \vdash U_1 \rhd a_1}\,\text{ind. hp.} \quad \ldots \quad \dfrac{\mathrm{G}_j(\mathbf{L}) \vdash U_n \rhd a_n}{\mathbf{L}' \vdash U_n \rhd a_n}\,\text{ind. hp.}}{\mathbf{L}' \vdash U \rhd a}\,r$$

(b) Case $r = \mathrm{L}j$ or r_j (where $r \in \mathcal{R}$): We have $a = j(b)$ for some $b \in S$, and hence we can write $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd j(b)$. Since $\mathrm{G}_j(\mathbf{L}) \subseteq \mathrm{M}_j(\mathbf{L}) \subseteq \mathrm{K}_j(\mathbf{L}')$, it follows that $\mathrm{K}_j(\mathbf{L}') \vdash U \rhd j(b)$. By the definition of $\mathrm{K}_j(\mathbf{L}')$, this is equivalent to $\mathbf{L}' \vdash U \rhd j(j(b))$. On the other hand, we observe that $\mathrm{M}_j(\mathbf{L}) \vdash j(j(b)) \rhd b$ by applying $\mathrm{L}j^+$ twice. Then $\mathrm{M}_j(\mathbf{L}) \subseteq \mathrm{K}_j(\mathbf{L}')$ implies $\mathrm{K}_j(\mathbf{L}') \vdash j(j(b)) \rhd b$, which is nothing but $\mathbf{L}' \vdash j(j(b)) \rhd j(b)$. By an application of transitivity between $\mathbf{L}' \vdash U \rhd j(j(b))$ and $\mathbf{L}' \vdash j(j(b)) \rhd j(b)$, we conclude $\mathbf{L}' \vdash U \rhd j(b)$. □

We can summarise the main situations in the following:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathbf{L}_1 \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathbf{L}_2 \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, equal] & \mathbf{L}_3 \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}_1) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j(\mathbf{L}_2) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}_3)
\end{tikzcd}$$

More detailed diagrams can be found in Appendix B.

3 Left and right nuclei

In the previous section, we introduced the notion of *bi-nucleus*, which allows rule $\mathrm{L}j$ to be applied to any element on the left-hand side of the sequent. This is sufficient for the main conservation theorem. However, in concrete cases many functions j satisfy only a weaker version of the rule $\mathrm{L}j$, which can be applied only to elements occurring in specific positions. For this reason, we introduce the notions of *left nucleus* and *right nucleus*.

Definition 4 ( ). *Consider the following rules:*

$$\frac{U, a \rhd j(b)}{U, j(a) \rhd j(b)}\,\mathrm{L}j_{\mathrm{left}} \qquad\qquad \frac{a, U \rhd j(b)}{j(a), U \rhd j(b)}\,\mathrm{L}j_{\mathrm{right}}$$

Given a set S endowed with an entailment relation $\mathbf{L}$, we say that an expansive function $j\colon S \to S$ is

- *a* right nucleus (over $\mathbf{L}$) *if* $\mathrm{L}j_{\mathrm{right}}$ *holds in* $\mathbf{L}$*;*
- *a* left nucleus (over $\mathbf{L}$) *if* $\mathrm{L}j_{\mathrm{left}}$ *holds in* $\mathbf{L}$*.*

Remark 2 ( ).

1. Every bi-nucleus is both a left and right nucleus.
2. If Comm is admissible in $\mathbf{L}$, then the definitions of right, left and bi-nucleus coincide.

Remark 3 ( ). Let j be a left/right nucleus over $\mathbf{L}$. Then an easy consequence of $\mathrm{L}j_*$ (where $* \in \{\mathrm{left}, \mathrm{right}\}$), the context-free version of transitivity survives after j:

$$\frac{a \rhd b \quad b \rhd c}{a \rhd c}\,\mathrm{trans} \qquad \rightsquigarrow \qquad \frac{a \rhd j(b) \quad \dfrac{b \rhd j(c)}{j(b) \rhd j(c)}\,\mathrm{L}j_*}{a \rhd j(c)}\,\mathrm{trans}$$

4 Full Lambek logic and its structural extensions

Fix a set S of formulae, which contains constants $0, 1$ and is closed under connectives $\wedge, \vee, \cdot, \multimapinv, \multimap$. As a basis, we consider *Full Lambek logic* **FL**, which is inductively defined by the rules in Table 1.

$$\frac{U, a, V \rhd c \quad U, b, V \rhd c}{U, a \vee b, V \rhd c}\,\mathrm{L}\vee \qquad \frac{U \rhd a}{U \rhd a \vee b}\,\mathrm{R}\vee_1 \qquad \frac{U \rhd b}{U \rhd a \vee b}\,\mathrm{R}\vee_2$$

$$\frac{U, a, V \rhd c}{U, a \wedge b, V \rhd c}\,\mathrm{L}\wedge_1 \qquad \frac{U, b, V \rhd c}{U, a \wedge b, V \rhd c}\,\mathrm{L}\wedge_2 \qquad \frac{U \rhd a \quad U \rhd b}{U \rhd a \wedge b}\,\mathrm{R}\wedge$$

$$\frac{U, V \rhd c}{U, 1, V \rhd c}\,\mathrm{L}1 \qquad \frac{}{\rhd 1}\,\mathrm{R}1$$

$$\frac{U, a, b, V \rhd c}{U, a \cdot b, V \rhd c}\,\mathrm{L}\cdot \qquad \frac{U \rhd a \quad V \rhd b}{U, V \rhd a \cdot b}\,\mathrm{R}\cdot$$

$$\frac{U \rhd a \quad W, b, V \rhd c}{W, U, a \multimap b, V \rhd c}\,\mathrm{L}\multimap \qquad \frac{a, U \rhd b}{U \rhd a \multimap b}\,\mathrm{R}\multimap$$

$$\frac{U \rhd a \quad W, b, V \rhd c}{W, b \multimapinv a, U, V \rhd c}\,\mathrm{L}\multimapinv \qquad \frac{U, a \rhd b}{U \rhd b \multimapinv a}\,\mathrm{R}\multimapinv$$

Table 1. The rules that generate **FL**, adapted from [11, §2.1]. To sometimes have a slicker notation, we subsume under L$\wedge$ and R$\vee$ the two left rules L$\wedge_1$ and L$\wedge_2$ and the two right rules R$\vee_1$ and R$\vee_2$, respectively.

Two unary connectives are defined as follows:

$$\sim a := a \multimap 0, \qquad\qquad -a := 0 \multimapinv a.$$

Remark 4 ( ). Let **L** be an inductive extension of **FL**.

1. If **L** admits Mono and Contr, then $\wedge$ and $\cdot$ coincide.
2. If **L** admits Comm, then $\multimapinv$ and $\multimap$ coincide.
3. R$\wedge$, R$\multimapinv$ and R$\multimap$ are invertible in **L**.

In **FL**+{Comm} and its extensions, we use the usual symbol $\to$ for symmetric implication and the usual symbol $\neg$ for symmetric negation.

An important structural extension is *Minimal logic*, which can be defined as $\mathbf{Min} := \mathbf{FL} + \{\text{Comm, Mono, Contr}\}$. We also define, as usual, *Intuitionistic logic* as $\mathbf{Int} := \mathbf{Min} + \{\mathrm{L0}\}$, where

$$\frac{}{U, 0, V \rhd c}\,\mathrm{L0}$$

For the rest of this section, let **L** be an inductive extension of **FL**, and let j be a function on formulae. By the *j-shift* for the logical operator $*$ we understand the corresponding axiom as in the following table:

$*$	**j-shift for $*$**
1	$1 \rhd j(1)$
$\wedge$	$j(a) \wedge j(b) \rhd j(a \wedge b)$
$\cdot$	$j(a) \cdot j(b) \rhd j(a \cdot b)$
$\vee$	$j(a) \vee j(b) \rhd j(a \vee b)$
$\multimap$	$a \multimap jb \rhd j(a \multimap b)$
$\circ\!\!-$	$a \circ\!\!- jb \rhd j(a \circ\!\!- b)$

Theorem 2 ( ). *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$*. The right rule* $\mathrm{R}*$ *is admissible for the Kleisli extension* $\mathrm{K}_j(\mathbf{L})$ *if and only if the* j*-shift for* $*$ *holds for* $\mathbf{L}$*.*

Proof. Let us consider the case $* = \wedge$; the other cases are similar.

Suppose first that the j-shift for $\wedge$ holds in $\mathbf{L}$. If $\mathrm{K}_j(\mathbf{L}) \vdash U \rhd a$ and $\mathrm{K}_j(\mathbf{L}) \vdash U \rhd b$, i.e. $\mathbf{L} \vdash U \rhd j(a)$ and $\mathbf{L} \vdash U \rhd j(b)$, then $\mathrm{R}\wedge$ for $\mathbf{L}$ gives $\mathbf{L} \vdash U \rhd j(a) \wedge j(b)$, which by transitivity and the j-shift for $\wedge$ implies $\mathbf{L} \vdash U \rhd j(a \wedge b)$, i.e. $\mathrm{K}_j(\mathbf{L}) \vdash U \rhd a \wedge b$.

As for the converse, assume that $\mathrm{R}\wedge$ is admissible in $\mathrm{K}_j(\mathbf{L})$. By reflexivity, $\mathbf{L} \vdash j(a) \rhd j(a)$ and $\mathbf{L} \vdash j(b) \rhd j(b)$. By $\mathrm{L}\wedge_1$ and $\mathrm{L}\wedge_2$, we get $\mathbf{L} \vdash j(a) \wedge j(b) \rhd j(a)$ and $\mathbf{L} \vdash j(a) \wedge j(b) \rhd j(b)$, i.e. $\mathrm{K}_j(\mathbf{L}) \vdash j(a) \wedge j(b) \rhd a$ and $\mathrm{K}_j(\mathbf{L}) \vdash j(a) \wedge j(b) \rhd b$. Then, $\mathrm{R}\wedge$ for $\mathrm{K}_j(\mathbf{L})$ entails $\mathrm{K}_j(\mathbf{L}) \vdash j(a) \wedge j(b) \rhd a \wedge b$, i.e. $\mathbf{L} \vdash j(a) \wedge j(b) \rhd j(a \wedge b)$. □

Lemma 1 ( ). *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$*, and let* j *be a (right / left / bi-) nucleus on* $\mathbf{L}$*. Then*

1. *The* j*-shifts for* 1*,* $\vee$ *hold in* $\mathbf{L}$*.*
2. *The* j*-shift for* $\cdot$ *holds in* $\mathbf{L} + \{\mathrm{Comm}\}$*.*
3. *The* j*-shift for* $\wedge$ *holds in* $\mathbf{L} + \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$*.*
4. *If the* j*-shift for* $\cdot$ *holds, then* j *is a bi-nucleus.*
5. *If* j *is a bi-nucleus, then* $\mathrm{L}\circ\!\!-$ *and* $\mathrm{L}\multimap$ *survive after* j*.*
6. *If the* j*-shift for* $\cdot$ *holds, then* $\mathrm{L}\circ\!\!-$*,* $\mathrm{L}\multimap$ *and* Trans *survive after* j*.*

Proof. Straightforward. For details, see Appendix A. □

Lemma 2 ( ). *Let* $\mathbf{L} = \mathbf{FL} + \mathcal{S} + \mathcal{R}$*, where* $\mathcal{S} \subseteq \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$*, and let* j *be a (right / left / bi-) nucleus on* $\mathbf{L}$*. Then*

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\cdot, j\text{-shift}\wedge, j\text{-shift}\circ\!\!-, j\text{-shift}\multimap\} + \mathcal{R}_j.$$

If, moreover, $\mathrm{Comm} \in \mathcal{S}$*, then*

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\wedge, j\text{-shift}\to\} + \mathcal{R}_j.$$

If, moreover, $\{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\} = \mathcal{S}$*, then*

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\to\} + \mathcal{R}_j.$$

Proof. By definition,

$$\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{\mathrm{L}_j,\ \mathrm{L1}_j,\ \mathrm{R1}_j,\ \mathrm{L}\vee_j,\ \mathrm{R}\vee_{1j},\ \mathrm{R}\vee_{2j},\ \mathrm{L}\wedge_{1j},\ \mathrm{L}\wedge_{2j},\ \mathrm{R}\wedge_j,\ \mathrm{L}\cdot_j,\ \mathrm{R}\cdot_j,\ \mathrm{L}\multimapinv_j,\ \mathrm{R}\multimapinv_j,\ \mathrm{L}\multimap_j,\ \mathrm{R}\multimap_j\} + \mathcal{S}_j + \mathcal{R}_j,$$

First, we check that all rules other than $\mathrm{R}\wedge$, $\mathrm{R}\multimapinv$, $\mathrm{R}\multimap$ survive after j. We have:

- It's straightforward that Rules Comm, Mono, Contr, if in $\mathcal{S}$, survive after j.
- Rules L1, $\mathrm{L}\vee$, $\mathrm{L}\wedge_1$, $\mathrm{L}\wedge_2$, $\mathrm{L}\cdot$ survive after j since they do not change succedents.
- Rules R1, $\mathrm{R}\vee_1$, $\mathrm{R}\vee_2$ survive after j by Theorem 2 since their j-shifts hold by Lemma 1.

Therefore we have $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{\mathrm{L}j, \mathrm{R}\cdot_j, \mathrm{R}\wedge_j, \mathrm{R}\multimapinv_j, \mathrm{R}\multimap_j, \mathrm{L}\multimap_j, \mathrm{L}\multimapinv_j\} + \mathcal{R}_j$. Observe that admissibilities of $\mathrm{R}\wedge_j$, $\mathrm{R}\cdot_j$, $\mathrm{R}\multimapinv_j$ and $\mathrm{R}\multimap_j$ are equivalent to admissibilities of j-shift$\wedge$, j-shift$\cdot$, j-shift$\multimapinv$, j-shift$\multimap$, respectively, by Theorem 2. By Lemma 1(6), we know that j-shift$\cdot$ suffices also for $\mathrm{L}\multimap_j$, $\mathrm{L}\multimapinv_j$ and Trans_j, which is tantamount to $\mathrm{L}j$ by Remark 1. Therefore we have $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\cdot, j\text{-shift}\wedge, j\text{-shift}\multimapinv, j\text{-shift}\multimap\} + \mathcal{R}_j$.

If, moreover, $\mathrm{Comm} \in \mathcal{S}$, then j-shift$\cdot$ holds by Lemma 1; and $\multimap$ and $\multimapinv$ coincide by Remark 4. Therefore we have $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\wedge, j\text{-shift}\to\} + \mathcal{R}_j$.

If, moreover, $\mathcal{S} = \{\mathrm{Comm}, \mathrm{Mono}, \mathrm{Contr}\}$, then j-shift$\wedge$ holds by Lemma 1. Therefore we conclude $\mathrm{G}_j(\mathbf{L}) = \mathbf{L} + \{j\text{-shift}\to\} + \mathcal{R}_j$. □

Theorem 3 (Conservation for FL). *Let j be a (right / left / bi-) nucleus over* **FL**, *and let* $\mathbf{L}$ *be an inductive extension of* **FL** *such that* $\mathbf{L} \subseteq \mathrm{M}_j(\mathbf{FL})$. *Then the following are equivalent:*

1. $\mathrm{M}_j(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd ja$.
2. $\mathrm{G}_j(\mathbf{FL}) \subseteq \mathbf{L}$.
3. *The following conditions hold:*
   - $\mathbf{L} \vdash ja \cdot jb \rhd j(a \cdot b)$,
   - $\mathbf{L} \vdash ja \wedge jb \rhd j(a \wedge b)$,
   - $\mathbf{L} \vdash a \multimap jb \rhd j(a \multimap b)$,
   - $\mathbf{L} \vdash jb \multimapinv a \rhd j(b \multimapinv a)$.

Proof. The equivalence between the first two items is an application of Theorem 1(4). The equivalence between the last two items is an application of Lemma 2. □

Remark 5 ( ). A common definition of a nucleus, as given by Johnstone [18, §C1.1], consists of the following conditions:

1. $a \rhd j(a)$,
2. $j(a) \lhd\!\rhd j(j(a))$,
3. $j(a \cdot b) \lhd\!\rhd j(a) \cdot j(b)$.

Conditions 1 and 2 follow directly from the rules Lj and Rj. Moreover, as a consequence of Lemma 2, we can observe that if all structural rules are satisfied, then condition 3 also holds. However, if Contr and Mono are omitted, condition 3 may not hold. Thus, in the substructural setting, these two notions of nucleus may differ.

In the final part of this section, we examine a few examples of nuclei and show how the general framework developed above allows us to recover a number of results from the literature. To keep the exposition focused on the main ideas, most of the technical proof details are omitted here; complete arguments can be found in Appendix A. We start by presenting two versions of the *Glivenko nucleus* [4,8], i.e. $a \mapsto \neg\neg a$, that work in substructural logic.

Remark 6 (🐌).

1. The *left Glivenko nucleus* $a \mapsto \sim\!-a$ is a left nucleus over **FL**;
2. The *right Glivenko nucleus* $a \mapsto -\!\sim a$ is a right nucleus over **FL**.

Corollary 1 (🐌).

1. *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$ *such that* $\mathbf{L} \subseteq \mathrm{M}_{\sim-}(\mathbf{FL})$. *Then the following are equivalent:*
   (a) $\mathrm{M}_{\sim-}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd \sim\!-a$.
   (b) $\mathrm{G}_{\sim-}(\mathbf{FL}) \subseteq \mathbf{L}$.
   (c) The following conditions hold:
   - $\mathbf{L} \vdash (\sim\!-a) \cdot (\sim\!-b) \rhd \sim\!-(a \cdot b)$,
   - $\mathbf{L} \vdash (\sim\!-a) \wedge (\sim\!-b) \rhd \sim\!-(a \wedge b)$,
   - $\mathbf{L} \vdash a \multimap \sim\!-b \rhd \sim\!-(a \multimap b)$,
   - $\mathbf{L} \vdash \sim\!-b \circ\!\!- a \rhd \sim\!-(b \circ\!\!- a)$,
2. *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$ *such that* $\mathbf{L} \subseteq \mathrm{M}_{-\sim}(\mathbf{FL})$. *Then the following are equivalent:*
   (a) $\mathrm{M}_{-\sim}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd -\!\sim a$.
   (b) $\mathrm{G}_{-\sim}(\mathbf{FL}) \subseteq \mathbf{L}$.
   (c) The following conditions hold:
   - $\mathbf{L} \vdash (-\!\sim a) \cdot (-\!\sim b) \rhd -\!\sim(a \cdot b)$,
   - $\mathbf{L} \vdash (-\!\sim a) \wedge (-\!\sim b) \rhd -\!\sim(a \wedge b)$,
   - $\mathbf{L} \vdash a \multimap -\!\sim b \rhd -\!\sim(a \multimap b)$,
   - $\mathbf{L} \vdash -\!\sim b \circ\!\!- a \rhd -\!\sim(b \circ\!\!- a)$,

Proof. Direct application of Theorem 3. □

If we consider the *Full Lambek logic with exchange* $\mathbf{FL_e} := \mathbf{FL} + \{\mathrm{Comm}\}$ [11, §2.1], then the *Glivenko nucleus* $\neg\neg$ is a nucleus over $\mathbf{FL_e}$.

We notice that the extension $\mathrm{M}_{\neg\neg}(\mathbf{FL_e}) = \mathbf{FL_e} + \{\neg\neg\text{-stab}\}$, corresponds to the one known in literature as the *Involutive Full Lambek logic with exchange* $\mathbf{InFL_e}$ [11, §7.3]. If, moreover, we consider **Min**, we obtain two well-known extensions: $\mathbf{Cla} := \mathrm{M}_{\neg\neg}(\mathbf{Min})$ is *Classical logic* and $\mathbf{Gli} := \mathrm{G}_{\neg\neg}(\mathbf{Min})$ is the *Glivenko logic* [11,28,34].

Corollary 2 (✍).

1. (Ono [29]) *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL_e}$ *such that* $\mathbf{L} \subseteq \mathbf{InFL_e}$. *Then the following are equivalent:*
   (a) $\mathbf{InFL_e} \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd \neg\neg a$.
   (b) $\mathrm{G}_{\neg\neg}(\mathbf{FL_e}) \subseteq \mathbf{L}$.
   (c) The following conditions hold:
   – $\mathbf{L} \vdash \neg\neg a \wedge \neg\neg b \rhd \neg\neg(a \wedge b)$,
   – $\mathbf{L} \vdash a \to \neg\neg b \rhd \neg\neg(a \to b)$.
2. (Odintsov [26,27])[6] *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{Min}$ *such that* $\mathbf{L} \subseteq \mathbf{Cla}$. *Then the following are equivalent:*
   (a) $\mathbf{Cla} \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U \rhd \neg\neg a$.
   (b) $\mathbf{Gli} \subseteq \mathbf{L}$.
   (c) $\mathbf{L} \vdash a \to \neg\neg b \rhd \neg\neg(a \to b)$.
3. (Glivenko [13,14]) $\mathbf{Cla} \vdash U \rhd a$ *if and only if* $\mathbf{Int} \vdash U \rhd \neg\neg a$.

Proof. Provided that $\mathbf{FL_e} \vdash (\neg\neg a) \cdot (\neg\neg b) \rhd \neg\neg(a \cdot b)$, these become particular cases of Corollary 2. The only thing left to show for item 3 is $\mathbf{Int} \vdash a \to \neg\neg b \rhd \neg\neg(a \to b)$, which is well-known (see, e.g. [10]). □

Lastly, we present a version of the *open nucleus* [4,8], i.e. $a \mapsto (0 \to a)$, that works in substructural logic.

Remark 7 (✍).

1. The *right open nucleus* $o_R : a \mapsto 0 \multimap 0 \cdot a$ is a right nucleus over $\mathbf{FL}$;
2. The *left open nucleus* $o_L : a \mapsto a \cdot 0 \multimapinv 0$ is a left nucleus over $\mathbf{FL}$.

Corollary 3 (✍).

1. *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$ *such that* $\mathbf{L} \subseteq \mathrm{M}_{o_R}(\mathbf{FL})$. *Then the following are equivalent:*
   (a) $\mathrm{M}_{o_R}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash 0, U \rhd 0 \cdot a$.
   (b) $\mathrm{G}_{o_R}(\mathbf{FL}) \subseteq \mathbf{L}$.
   (c) The following conditions hold:
   – $\mathbf{L} \vdash 0, (0 \multimap 0 \cdot a) \cdot (0 \multimap 0 \cdot b) \rhd 0 \cdot (a \cdot b)$,
   – $\mathbf{L} \vdash 0, (0 \multimap 0 \cdot a) \wedge (0 \multimap 0 \cdot b) \rhd 0 \cdot (a \wedge b)$,
   – $\mathbf{L} \vdash 0, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot (a \multimap b)$,
   – $\mathbf{L} \vdash 0, (0 \multimap 0 \cdot b) \multimapinv a \rhd 0 \cdot (b \multimapinv a)$,
2. *Let* $\mathbf{L}$ *be an inductive extension of* $\mathbf{FL}$ *such that* $\mathbf{L} \subseteq \mathrm{M}_{o_L}(\mathbf{FL})$. *Then the following are equivalent:*
   (a) $\mathrm{M}_{o_L}(\mathbf{FL}) \vdash U \rhd a$ *if and only if* $\mathbf{L} \vdash U, 0 \rhd a \cdot 0$.
   (b) $\mathrm{G}_{o_L}(\mathbf{FL}) \subseteq \mathbf{L}$.
   (c) The following conditions hold:

[6] This was proved by Odintsov [26] and shortly after by Cignoli and Torres [7]. This fact was already noted by Segerberg, who mentioned it on [34, page 46] without providing any proof or further reference. In fact, the first published proofs that we are aware of are the two mentioned above.

- $\mathbf{L} \vdash (a \cdot 0 \multimapinv 0) \cdot (b \cdot 0 \multimapinv 0), 0 \rhd (a \cdot b) \cdot 0$,
- $\mathbf{L} \vdash (a \cdot 0 \multimapinv 0) \wedge (b \cdot 0 \multimapinv 0), 0 \rhd (a \wedge b) \cdot 0$,
- $\mathbf{L} \vdash a \multimap (b \cdot 0 \multimapinv 0), 0 \rhd (a \multimap b) \cdot 0$,
- $\mathbf{L} \vdash (b \cdot 0 \multimapinv 0) \multimapinv a, 0 \rhd (b \multimapinv a) \cdot 0$,

Proof. This is Theorem 3 with invertibility of R$\multimap$ and R$\multimapinv$ (Remark 4). □

5 Future work

In this paper, we have focused on the provability-theoretic aspects of Glivenko-type theorems and j-shift admissibility within substructural entailment relations. However, our approach has remained *proof-irrelevant* in that it treats derivability relations extensionally, without distinguishing between different proofs of the same judgment.

A natural direction for future work is to explore the same framework from a *proof-relevant* perspective. This would involve enriching the underlying entailment relations to retain information about the structure or identity of proofs, such as labelling derivations, formalising proof terms, or working within a categorical semantics where morphisms represent proofs explicitly.

Several questions arise in this setting:

- How do Glivenko extensions interact with proof equivalences? Given an equivalence relation on proofs in an entailment relation $\mathbf{L}$, can one define a canonical transformation between proofs of $\mathbf{L} \vdash U \rhd a$ and proofs of $\mathrm{G}_j(\mathbf{L}) \vdash U \rhd j(a)$ in a way that preserves the equivalence relation? Moreover, if such a transformation exists, does it have an inverse? This question is related to the decorations studied in Schellinx's PhD thesis [33], where the codomain of the translation is classical linear logic, while here we are in a more abstract setting.
- In what sense are j-shifts invertible or conservative at the level of proofs, not just derivability? What kinds of monads arise categorically?
- Can we extend our results to a setting like type theory or logical frameworks (e.g., LF [15], FOLDS [23]), where proofs themselves are first-class objects?

Adopting a proof-relevant viewpoint could also facilitate connections with categorical logic and type theory. Additionally, this perspective might yield new insights into normalisation, proof transformation, or computational interpretations of substructural logics. We leave these directions for future investigation.

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References

1. Aczel, P.: The Russell–Prawitz modality. Math. Struct. Comput. Sci. **11**(4), 541–554 (2001). https://doi.org/10.1017/s0960129501003309
2. Aczel, P.: Aspects of general topology in constructive set theory. Ann. Pure Appl. Logic **137**(1–3), 3–29 (2006). https://doi.org/10.1016/j.apal.2005.05.016
3. Altenkirch, T.: Nuclear — Agda formalisation. https://github.com/txa/nuclear (2024)
4. Berg, B.v.d.: A Kuroda-style j-translation. Arch. Math. Log. **58**(5), 627–634 (2019). https://doi.org/10.1007/s00153-018-0656-x
5. Borsetto, R.: Glivenko's theorem underneath structure — Agda formalisation. https://github.com/eapiova/nuclear (2026)
6. Cignoli, R., Torrens, A.: Hájek basic fuzzy logic and Lukasiewicz infinite-valued logic. Arch. Math. Log. **42**(4), 361–370 (2003). https://doi.org/10.1007/s001530200144
7. Cignoli, R., Torrens, A.: Glivenko like theorems in natural expansions of BCK-logic. Math. Log. Q. **50**, 111–125 (2004). https://doi.org/10.1002/malq.200310082
8. Escardó, M., Oliva, P.: The Peirce translation. Ann. Pure Appl. Logic **163**, 681–692 (2012). https://doi.org/10.1016/j.apal.2011.11.002
9. Fellin, G., Schuster, P.: A general Glivenko–Gödel theorem for nuclei. In: Sokolova, A. (ed.) Proc. of 37th Conf. on the Mathematical Foundations of Programming Semantics, MFPS 2021, Electron. Proc. in Theor. Comput. Sci., vol. 351, pp. 51–66. Open Publishing Assoc. (2021). https://doi.org/10.4204/eptcs.351.4
10. Fellin, G., Schuster, P.: Conservation as translation. Rev. Symb. Log. **18**(1), 316–348 (2025). https://doi.org/10.1017/s1755020324000170
11. Galatos, N., Jipsen, P., Kowalski, T., Ono, H.: Residuated Lattices: An Algebraic Glimpse at Substructural Logics, Studies in Logic and the Foundations of Mathematics, vol. 151. Elsevier (2007)
12. Galatos, N., Ono, H.: Glivenko theorems for substructural logics over FL. J. Symb. Log. **71**(4), 1353–1384 (2006). https://doi.org/10.2178/jsl/1164060460
13. Glivenko, V.: Sur la logique de M. Brouwer. Acad. Roy. Belg. Bull. Cl. Sci. (5) **14**, 225–228 (1928)
14. Glivenko, V.: Sur quelques points de la logique de M. Brouwer. Acad. Roy. Belg. Bull. Cl. Sci. (5) **15**, 183–188 (1929)
15. Harper, R., Honsell, F., Plotkin, G.D.: A framework for defining logics. J. ACM **40**(1), 143–184 (1993). https://doi.org/10.1145/138027.138060
16. Haykazyan, L.: More on a curious nucleus. J. Pure Appl. Algebra **224**, 860–868 (2020). https://doi.org/10.1016/j.jpaa.2019.06.014
17. Johnstone, P.T.: Stone Spaces, Cambridge Studies in Advanced Mathematics, vol. 3. Cambridge University Press (1982)
18. Johnstone, P.T.: Sketches of an Elephant: A Topos Theory Compendium. Vol. 2, Oxford Logic Guides, vol. 44. Clarendon Press (2002)
19. Kock, A.: Synthetic Differential Geometry, London Math. Soc. Lecture Note Ser., vol. 333. Cambridge University Press, 2nd edn. (2006). https://doi.org/10.1017/cbo9780511550812
20. Lambek, J.: The mathematics of sentence structure. American Mathematical Monthly **65**, 154–170 (1958), https://api.semanticscholar.org/CorpusID:123801856
21. Lambek, J.: On the calculus of syntactic types. In: Jakobson, R. (ed.) Structure of Language and its Mathematical Aspects (1961), https://api.semanticscholar.org/CorpusID:118284222

22. Lambek, J.: Some lattice models of bilinear logic. algebra universalis **34**, 541–550 (1995), https://api.semanticscholar.org/CorpusID:119782781
23. Makkai, M.: Towards a categorical foundation of mathematics. In: Makowsky, J.A., Ravve, E.V. (eds.) Logic Colloquium '95. Lecture Notes in Logic, vol. 11, pp. 153–190. Cambridge University Press (1998)
24. Negri, S.: Continuous domains as formal spaces. Math. Struct. Comput. Sci. **12**(1), 19–52 (2002). https://doi.org/10.1017/s0960129501003450
25. Norell, U.: Towards a Practical Programming Language Based on Dependent Type Theory. Ph.D. thesis, Chalmers University of Technology (2007)
26. Odintsov, S.P.: Logic of classical refutability and class of extensions of minimal logic. Logic and Logical Philosophy **9**(9), 91–107 (2001). https://doi.org/10.12775/llp.2001.006
27. Odintsov, S.P.: Negative equivalence of extensions of minimal logic. Studia Logica **78**(3), 417–442 (2004). https://doi.org/10.1007/s11225-004-6043-0
28. Odintsov, S.P.: Constructive Negations and Paraconsistency, Trends in Logic, vol. 26. Kluwer (2008). https://doi.org/10.1007/978-1-4020-6867-6
29. Ono, H.: Glivenko theorems revisited. Ann. Pure Appl. Log. **161**(2), 246–250 (2009). https://doi.org/10.1016/j.apal.2009.05.006
30. Paoli, F.: Substructural Logics: A Primer, Trends in Logic, vol. 13. Kluwer (2002). https://doi.org/10.1007/978-94-017-3179-9
31. Restall, G.: Substructural Logics. In: Zalta, E.N., Nodelman, U. (eds.) The Stanford Encyclopedia of Philosophy. Metaphysics Research Lab, Stanford University, Fall 2024 edn. (2024)
32. Rosenthal, K.I.: Quantales and their Applications, Pitman Research Notes in Mathematics, vol. 234. Longman Scientific & Technical, Essex (1990)
33. Schellinx, H.: The Noble Art of Linear Decorating. Ph.D. thesis, University of Amsterdam (1994), https://eprints.illc.uva.nl/id/eprint/1964
34. Segerberg, K.: Propositional logics related to Heyting's and Johansson's. Theoria **34**(1), 26–61 (1968). https://doi.org/10.1111/j.1755-2567.1968.tb00337.x
35. Simmons, H.: A framework for topology. In: Macintyre, A., Pacholski, L., Paris, J. (eds.) Logic Colloquium '77, Studies in Logic and the Foundations of Mathematics, vol. 96, pp. 239–251. North-Holland, Amsterdam (1978). https://doi.org/10.1016/S0049-237x(08)72007-x
36. Simmons, H.: A curious nucleus. J. Pure Appl. Algebra **214**, 2063–2073 (2010). https://doi.org/10.1016/j.jpaa.2010.02.011
37. Vezzosi, A., Mörtberg, A., Abel, A.: Cubical Agda: A dependently typed programming language with univalence and higher inductive types. Journal of Functional Programming **31** (2021), e8

Appendix

A Proof details

Proof of Remark 1.

1. A conclusion $a \rhd j(a)$ of Refl_j can be derived by subsequent applications of Refl and Rj. On the other hand, if we have the premiss $U \rhd b$ of Rj and by Refl_j we can derive $b \rhd j(b)$, then by Trans we can derive $U \rhd j(b)$, which is the conclusion of Rj.
2. Suppose that we have the premisses $V_1, a, V_2 \rhd j(b)$ and $V_1, j(a), V_2 \rhd j(b)$ of Trans_j. If we apply Lj to the latter, and then apply Trans with the former, we derive $V_1, U, V_2 \rhd j(b)$, which is the conclusion of Trans_j. On the other hand, if we have the premiss $U, a, V \rhd j(b)$ of Lj, we can apply Trans_j to it with $j(a) \rhd j(a)$ to obtain $U, j(a), V \rhd j(b)$, which is the conclusion of Lj.
3. Similar to the the first item. □

We recall a couple of well-known results whose proofs are straightforward:

Remark 8 (🤓). The following hold in **FL**:

$$\frac{}{a, a \multimap b \rhd b}\,\mathrm{mp}{\multimap} \qquad\qquad \frac{}{b \multimapinv a, a \rhd b}\,\mathrm{mp}{\multimapinv}$$

Proof of Lemma 1.

1. The j-shift for 1 is $1 \rhd j(1)$, which holds in in **L** as a direct consequence of Rj. We prove that j-shift$\vee_1$ holds in **L**:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{a \rhd a}\,\mathrm{Refl}}{a \rhd a \vee b}\,\mathrm{R}\vee_1}{a \rhd j(a \vee b)}\,\mathrm{R}j}{j(a) \rhd j(a \vee b)}\,\mathrm{L}j_* \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{b \rhd b}\,\mathrm{Refl}}{b \rhd a \vee b}\,\mathrm{R}\vee_2}{b \rhd j(a \vee b)}\,\mathrm{R}j}{j(b) \rhd j(a \vee b)}\,\mathrm{L}j_*}{j(a) \vee j(b) \rhd j(a \vee b)}\,\mathrm{L}\vee$$

   where Lj_* is either Lj_{right} or Lj_{left}.
2. We prove that j-shift$\cdot$ holds in $\mathbf{L} + \{\mathrm{Comm}\}$ in the case of j being a right nucleus:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{a \rhd a}\,\mathrm{Refl} \quad \dfrac{}{b \rhd b}\,\mathrm{Refl}}{a, b \rhd a \cdot b}\,\mathrm{R}\cdot}{a, b \rhd j(a \cdot b)}\,\mathrm{R}j}{b, a \rhd j(a \cdot b)}\,\mathrm{Comm}}{j(b), a \rhd j(a \cdot b)}\,\mathrm{L}j_{\mathrm{right}}}{a, j(b) \rhd j(a \cdot b)}\,\mathrm{Comm}}{j(a), j(b) \rhd j(a \cdot b)}\,\mathrm{L}j_{\mathrm{right}}}{j(a) \cdot j(b) \rhd j(a \cdot b)}\,\mathrm{L}\cdot$$

   The case involving a left nucleus is similar.

3. Since j-shift$\cdot$ holds in $\mathbf{L} + \{\text{Comm}\}$, it holds in $\mathbf{L} + \{\text{Comm, Mono, Contr}\}$ as well. Then j-shift$\wedge$ holds in $\mathbf{L} + \{\text{Comm, Mono, Contr}\}$ by Remark 4.
4. Let us prove the case in which j is left, the other is similar. Denote by $\bullet V$ the formula $v_1 \cdot ... \cdot v_n$, where $V = v_1, ..., v_n$. We show that Lj holds:

$$\dfrac{\dfrac{\dfrac{}{j(a) \rhd j(a)}\,\text{Refl} \quad \dfrac{\dfrac{}{V \rhd \bullet V}\,(*)}{V \rhd j(\bullet V)}\,\text{R}j}{j(a), V \rhd j(a) \cdot j(\bullet V)}\,\text{R}\wedge \quad \dfrac{\dfrac{}{j(a) \cdot j(\bullet V) \rhd j(a \cdot \bullet V)}\,j\text{-shift}\cdot \quad \dfrac{\dfrac{U, a, V \rhd j(b)}{U, a \cdot \bullet V \rhd j(b)}\,(**)}{U, j(a \cdot \bullet V) \rhd j(b)}\,\text{L}j_{\text{left}}}{U, j(a) \cdot j(\bullet V) \rhd j(b)}\,\text{Trans}}{U, j(a), V \rhd j(b)}\,\text{Trans}$$

where steps $(*)$ and $(**)$ correspond to several subsequent applications of R$\cdot$ and L$\cdot$, respectively, and are straightforwardly provable by induction.
5. Let us prove L$\multimap_j$, the other case is similar.

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{a, a \multimap b \rhd b}\,\text{mp}\multimap}{a, a \multimap b \rhd j(b)}\,\text{R}j}{j(a), a \multimap b \rhd j(b)}\,\text{L}j}{a \multimap b \rhd j(a) \multimap j(b)}\,\text{R}\multimap \quad \dfrac{U \rhd j(a) \quad \dfrac{W, b, V \rhd j(c)}{W, j(b), V \rhd j(c)}\,\text{L}j}{W, U, j(a) \multimap j(b), V \rhd j(c)}\,\text{L}\multimap}{W, U, a \multimap b, V \rhd j(c)}\,\text{Trans}$$

6. Follows from the previous two items and Remark 3. □

Proof of Remark 6. We only give the proof for L$\sim-_{\text{left}}$ R$\sim-$; the proof for $-\sim$ is similar.

$$\dfrac{\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\text{Refl} \quad \dfrac{\dfrac{U, a \rhd \sim -b \quad \dfrac{\dfrac{}{0 \rhd 0}\,\text{Refl} \quad \dfrac{}{-b \rhd -b}\,\text{Refl}}{-b, \sim -b \rhd 0}\,\text{L}\multimap}{-b, U, a \rhd 0}\,\text{Trans}}{-b, U \rhd -a}\,\text{R}\multimapinv}{-b, U, \sim -a \rhd 0}\,\text{L}\multimap}{U, \sim -a \rhd \sim -b}\,\text{R}\multimap \qquad \dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\text{Refl} \quad U \rhd a}{-a, U \rhd 0}\,\text{L}\multimapinv}{U \rhd \sim -a}\,\text{R}\multimap$$

□

Proof of Corollary 2. We prove that $\mathbf{FL} + \text{Comm} \vdash \sim-$-shift$\cdot$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{a \rhd a}\,\text{Refl} \quad \dfrac{}{b \rhd b}\,\text{Refl}}{a, b \rhd a \cdot b}\,\text{R}\cdot}{b, a \rhd a \cdot b}\,\text{Comm} \quad \dfrac{}{0 \rhd 0}\,\text{Refl}}{-(a \cdot b), b, a \rhd 0}\,\text{L}\multimapinv}{-(a \cdot b), b \rhd -a}\,\text{R}\multimapinv \quad \dfrac{}{0 \rhd 0}\,\text{Refl}}{-(a \cdot b), b, \sim -a \rhd 0}\,\text{L}\multimap}{-(a \cdot b), \sim -a, b \rhd 0}\,\text{Comm}}{-(a \cdot b), \sim -a \rhd -b}\,\text{R}\multimapinv \quad \dfrac{}{0 \rhd 0}\,\text{Refl}}{-(a \cdot b), \sim -a, \sim -b \rhd 0}\,\text{L}\multimap}{\sim -a, \sim -b \rhd \sim -(a \cdot b)}\,\text{R}\multimap}{\sim -a \cdot \sim -b \rhd \sim -(a \cdot b)}\,\text{L}\cdot$$

□

Proof of Remark 7. We only give the prove for o_R, the proof for o_L is similar. About $\mathrm{R}o_R$:

$$\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \qquad U \rhd a}{0, U \rhd 0 \cdot a}\,\mathrm{R}\cdot}{U \rhd 0 \multimap 0 \cdot a}\,\mathrm{R}\multimap$$

Before proving $\mathrm{L}o_{R\mathrm{right}}$, we need:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \qquad \dfrac{\dfrac{}{a \rhd a}\,\mathrm{Refl} \qquad \dfrac{\dfrac{}{0 \rhd 0}\,\mathrm{Refl} \qquad \dfrac{}{0 \cdot b \rhd 0 \cdot b}\,\mathrm{Refl}}{0, 0 \multimap 0 \cdot b \rhd 0 \cdot b}\,\mathrm{L}\multimap}{0, a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot b}\,\mathrm{L}\multimap}{0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot b}\,\mathrm{L}\cdot}{0, 0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \cdot b}\,\mathrm{L}\multimap}{0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}\,\mathrm{R}\multimap}{a \multimap (0 \multimap 0 \cdot b) \rhd (0 \multimap 0 \cdot a) \multimap (0 \multimap 0 \cdot b)}\,\mathrm{R}\multimap \qquad \dfrac{\dfrac{}{0 \multimap 0 \cdot a \rhd 0 \multimap 0 \cdot a}\,\mathrm{Refl} \qquad \dfrac{}{0 \multimap 0 \cdot b \rhd 0 \multimap 0 \cdot b}\,\mathrm{Refl}}{0 \multimap 0 \cdot a, (0 \multimap 0 \cdot a) \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}\,\mathrm{L}\multimap}{0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}\,\mathrm{Trans}$$

We can now prove $\mathrm{L}o_{R\mathrm{right}}$:

$$\dfrac{\dfrac{a, U \rhd 0 \multimap 0 \cdot b}{U \rhd a \multimap (0 \multimap 0 \cdot b)}\,\mathrm{R}\multimap \qquad \dfrac{}{0 \multimap 0 \cdot a, a \multimap (0 \multimap 0 \cdot b) \rhd 0 \multimap 0 \cdot b}}{0 \multimap 0 \cdot a, U \rhd 0 \multimap 0 \cdot b}\,\mathrm{Trans}$$

□

B Diagrams of extensions

Supposing that $L \subsetneq \mathrm{G}_j(\mathbf{L}) \subsetneq \mathrm{M}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L})
\end{tikzcd}$$

We want to see what happens if we add some inductive extension $\mathbf{L}'$ of $\mathbf{L}$ to the diagram.

1. Case $\mathbf{L}'$ unrelated to $\mathrm{M}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

2. Case $\mathbf{L}' \subsetneq \mathrm{M}_j(\mathbf{L})$ but unrelated to $\mathrm{G}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[dr, hook] \arrow[dd, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[dr, hook] \arrow[dd, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[r, hook] \arrow[ur, hook] \arrow[dd, hook] & \mathrm{G}_j(\mathbf{L}') \arrow[r, hook] \arrow[dd, hook] & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] \arrow[dr, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] \arrow[dr, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

3. Case $\mathbf{L} \subsetneq \mathbf{L}' \subsetneq \mathrm{G}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[dr, hook] \arrow[dd, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[dr, equal] \arrow[dd, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[r, hook] \arrow[ur, hook] \arrow[dd, hook] & \mathrm{G}_j(\mathbf{L}') \arrow[r, hook] \arrow[dd, hook] & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] \arrow[dr, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] \arrow[dr, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathbf{L}' \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}') \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j(\mathbf{L}') \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}') \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

4. Case $\mathrm{G}_j(\mathbf{L}) \subsetneq \mathbf{L}' \subsetneq \mathrm{M}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[dr, hook] \arrow[dd, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[dr, hook] \arrow[dl] \arrow[dd, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[r, equal] \arrow[dd, hook] & \mathrm{G}_j(\mathbf{L}') \arrow[r, hook] \arrow[dd, hook] & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] \arrow[dr, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] \arrow[dr, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathbf{L}' \arrow[r, equal] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}') \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j(\mathbf{L}') \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

5. Case $\mathrm{M}_j(\mathbf{L}) \subsetneq \mathbf{L}'$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] \arrow[dd, equal] \arrow[dll, hook] & \\
& \mathbf{L}' \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, hook] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, equal] & \mathbf{L}' \arrow[r, equal] \arrow[d, equal] & \mathrm{G}_j(\mathbf{L}') \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j(\mathbf{L}') \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

6. Case $\mathbf{L}' = \mathbf{L}$: Trivial.

7. Case $\mathbf{L}' = \mathrm{G}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] \arrow[dl, equal] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] & \\
& \mathbf{L}' \arrow[rr, equal] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, hook] & \mathrm{G}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j\,\mathrm{G}_j(\mathbf{L})
\end{tikzcd}$$

8. Case $\mathbf{L}' = \mathrm{M}_j(\mathbf{L})$:

$$\begin{tikzcd}
\mathbf{L} \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{G}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] \arrow[dd, hook] & & \mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] \arrow[dd, equal] \arrow[dll, equal] & \\
& \mathbf{L}' \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] \arrow[dd, equal] & & \mathrm{M}_j(\mathbf{L}') \arrow[dd, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[rr, hook] \arrow[dr, hook] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[rr, equal] \arrow[dr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[dr, equal] & \\
& \mathrm{K}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}') \arrow[rr, equal] & & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}')
\end{tikzcd}$$

Simplified:

$$\begin{tikzcd}
\mathbf{L} \arrow[r, hook] \arrow[d, hook] & \mathrm{G}_j(\mathbf{L}) \arrow[r, hook] \arrow[d, hook] & \mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{G}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] \arrow[d, equal] & \mathrm{M}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[d, equal] \\
\mathrm{K}_j(\mathbf{L}) \arrow[r, hook] & \mathrm{K}_j\,\mathrm{G}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{G}_j\,\mathrm{M}_j(\mathbf{L}) \arrow[r, equal] & \mathrm{K}_j\,\mathrm{M}_j\,\mathrm{M}_j(\mathbf{L})
\end{tikzcd}$$